In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import statsmodels.api as sm
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
import zipfile
from scipy import stats
```

In [2]:
```
zip_file_path = "C:\\Users\\PC\\Documents\\UoM\\BMAN73701 - Python\\Coursework\\Produc
```

In [3]:
```
extracted_dir = 'C:\\Users\\PC\\Documents\\UoM\\BMAN73701 - Python\\Coursework'
```

In [4]:
```
with zipfile.ZipFile(zip_file_path, 'r') as zip_ref:
    # Extract all contents to the specified directory
    zip_ref.extractall(extracted_dir)
```

In [5]:
```
file = "Products_Information.csv"
```

In [6]:
```
#1.1. Load the data
data = pd.read_csv(file)
data.head(-10)
```

Out[6]:

	id	date	store_nbr	product_type	sales	special_offer
0	0	2013-01-01	1	AUTOMOTIVE	0.0	0
1	1	2013-01-01	1	BABY CARE	0.0	0
2	2	2013-01-01	1	BEAUTY	0.0	0
3	3	2013-01-01	1	BEVERAGES	0.0	0
4	4	2013-01-01	1	BOOKS	0.0	0
...	...	...	...	...	...	...
3000873	3000873	2017-08-15	9	HOME CARE	373.0	6
3000874	3000874	2017-08-15	9	LADIESWEAR	10.0	0
3000875	3000875	2017-08-15	9	LAWN AND GARDEN	11.0	3
3000876	3000876	2017-08-15	9	LINGERIE	6.0	0
3000877	3000877	2017-08-15	9	LIQUOR,WINE,BEER	25.0	2

3000878 rows × 6 columns

In [7]:
```
#1.2. Exlore the data
data.info()
data.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000888 entries, 0 to 3000887
Data columns (total 6 columns):
 #   Column         Dtype
---  ------         -----
 0   id             int64
 1   date           object
 2   store_nbr      int64
 3   product_type   object
 4   sales          float64
 5   special_offer  int64
dtypes: float64(1), int64(3), object(2)
memory usage: 137.4+ MB
```

Out[7]:

	id	store_nbr	sales	special_offer
count	3.000888e+06	3.000888e+06	3.000888e+06	3.000888e+06
mean	1.500444e+06	2.750000e+01	3.577757e+02	2.602770e+00
std	8.662819e+05	1.558579e+01	1.101998e+03	1.221888e+01
min	0.000000e+00	1.000000e+00	0.000000e+00	0.000000e+00
25%	7.502218e+05	1.400000e+01	0.000000e+00	0.000000e+00
50%	1.500444e+06	2.750000e+01	1.100000e+01	0.000000e+00
75%	2.250665e+06	4.100000e+01	1.958473e+02	0.000000e+00
max	3.000887e+06	5.400000e+01	1.247170e+05	7.410000e+02

In [8]:

```python
#1.3. Observe the products and stores that need to be forecasted
unique_products = data['product_type'].unique()
print(unique_products)
unique_stores = data['store_nbr'].unique()
print(unique_stores)
```

```
['AUTOMOTIVE' 'BABY CARE' 'BEAUTY' 'BEVERAGES' 'BOOKS' 'BREAD/BAKERY'
 'CELEBRATION' 'CLEANING' 'DAIRY' 'DELI' 'EGGS' 'FROZEN FOODS' 'GROCERY I'
 'GROCERY II' 'HARDWARE' 'HOME AND KITCHEN I' 'HOME AND KITCHEN II'
 'HOME APPLIANCES' 'HOME CARE' 'LADIESWEAR' 'LAWN AND GARDEN' 'LINGERIE'
 'LIQUOR,WINE,BEER' 'MAGAZINES' 'MEATS' 'PERSONAL CARE' 'PET SUPPLIES'
 'PLAYERS AND ELECTRONICS' 'POULTRY' 'PREPARED FOODS' 'PRODUCE'
 'SCHOOL AND OFFICE SUPPLIES' 'SEAFOOD']
[ 1 10 11 12 13 14 15 16 17 18 19  2 20 21 22 23 24 25 26 27 28 29  3 30
 31 32 33 34 35 36 37 38 39  4 40 41 42 43 44 45 46 47 48 49  5 50 51 52
 53 54  6  7  8  9]
```

In [9]:

```python
#1.4. Check for missing values
print("Number of missing values:\n", data.isnull().sum())

# Handle missing values (e.g., forward fill, backward fill, interpolation)
data['sales'].fillna(method='ffill', inplace=True)
```

```
Number of missing values:
 id               0
date             0
store_nbr        0
product_type     0
sales            0
special_offer    0
dtype: int64
```

In [10]:

```
#1.5 Extract the data for 2016 for training purpose
data.index = pd.to_datetime(data['date'])
data['date'] = pd.to_datetime(data['date'])

data_train = data.loc['2016',].copy()
data_train
```

Out[10]:

	id	date	store_nbr	product_type	sales	special_offer
date						
2016-01-01	1945944	2016-01-01	1	AUTOMOTIVE	0.000	0
2016-01-01	1945945	2016-01-01	1	BABY CARE	0.000	0
2016-01-01	1945946	2016-01-01	1	BEAUTY	0.000	0
2016-01-01	1945947	2016-01-01	1	BEVERAGES	0.000	0
2016-01-01	1945948	2016-01-01	1	BOOKS	0.000	0
...	...	...	...	...	...	...
2016-12-31	2596369	2016-12-31	9	POULTRY	687.853	1
2016-12-31	2596370	2016-12-31	9	PREPARED FOODS	100.405	1
2016-12-31	2596371	2016-12-31	9	PRODUCE	3091.356	3
2016-12-31	2596372	2016-12-31	9	SCHOOL AND OFFICE SUPPLIES	2.000	0
2016-12-31	2596373	2016-12-31	9	SEAFOOD	13.000	2

650430 rows × 6 columns

In [11]:

```
#1.6. Extract data for testing purpose
start_date = '2017-07-17'
end_date = '2017-08-15'
data_test =  data[(data['date'] >= start_date) & (data['date'] <= end_date)]
data_test
```

Out[11]:

	id	date	store_nbr	product_type	sales	special_offer
date						
2017-07-17	2947428	2017-07-17	1	AUTOMOTIVE	2.000	0
2017-07-17	2947429	2017-07-17	1	BABY CARE	0.000	0
2017-07-17	2947430	2017-07-17	1	BEAUTY	5.000	0
2017-07-17	2947431	2017-07-17	1	BEVERAGES	2381.000	39
2017-07-17	2947432	2017-07-17	1	BOOKS	1.000	0
...	...	...	...	...	...	...
2017-08-15	3000883	2017-08-15	9	POULTRY	438.133	0
2017-08-15	3000884	2017-08-15	9	PREPARED FOODS	154.553	1
2017-08-15	3000885	2017-08-15	9	PRODUCE	2419.729	148
2017-08-15	3000886	2017-08-15	9	SCHOOL AND OFFICE SUPPLIES	121.000	8
2017-08-15	3000887	2017-08-15	9	SEAFOOD	16.000	0

53460 rows × 6 columns

In [12]:

```
#1.7. Check for outliers

# Determine z-score for sales data
z_scores = stats.zscore(data_train['sales'])
threshold = 3

# Identify outliers
outliers = (np.abs(z_scores) > threshold)

# Display or filter outliers
print(data_train[outliers])
```

```
                 id       date  store_nbr product_type      sales  \
date
2016-01-01  1946508 2016-01-01         25    BEVERAGES   5104.000
2016-01-02  1947795 2016-01-02         11    BEVERAGES   8238.000
2016-01-02  1947804 2016-01-02         11    GROCERY I  10279.158
2016-01-02  1947870 2016-01-02         13    GROCERY I   5488.000
2016-01-02  1947903 2016-01-02         14    GROCERY I   4354.000
...             ...        ...        ...          ...        ...
2016-12-31  2596311 2016-12-31          8    BEVERAGES   5596.000
2016-12-31  2596320 2016-12-31          8    GROCERY I   5317.000
2016-12-31  2596338 2016-12-31          8      PRODUCE   5993.142
2016-12-31  2596344 2016-12-31          9    BEVERAGES   4435.000
2016-12-31  2596353 2016-12-31          9    GROCERY I   7657.226

            special_offer
date
2016-01-01              1
2016-01-02              2
2016-01-02              6
2016-01-02              7
2016-01-02              1
...                   ...
2016-12-31             66
2016-12-31            113
2016-12-31              8
2016-12-31             67
2016-12-31            147

[14111 rows x 6 columns]
```

In [13]:

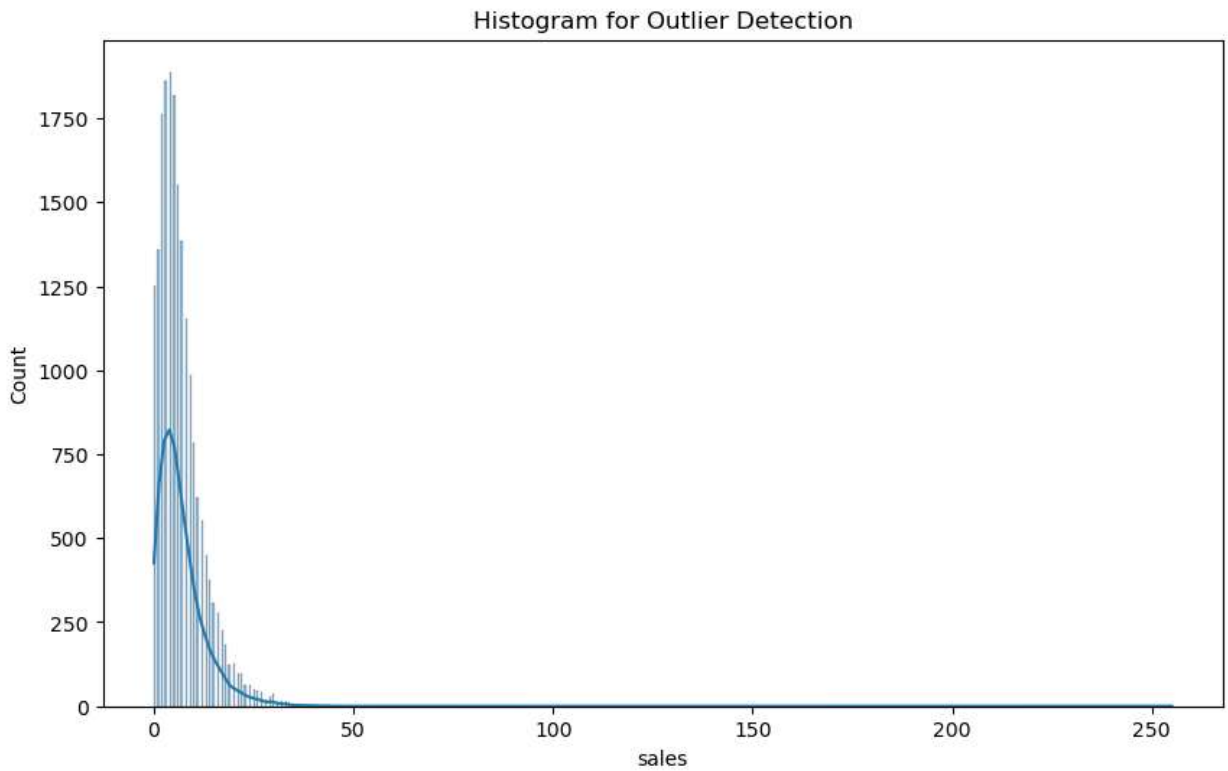
```
#Extract data for only automotive
automotive_train = data_train[data_train['product_type'] == "AUTOMOTIVE"]
automotive_train
```

Out[13]:

	id	date	store_nbr	product_type	sales	special_offer
date						
2016-01-01	1945944	2016-01-01	1	AUTOMOTIVE	0.0	0
2016-01-01	1945977	2016-01-01	10	AUTOMOTIVE	0.0	0
2016-01-01	1946010	2016-01-01	11	AUTOMOTIVE	0.0	0
2016-01-01	1946043	2016-01-01	12	AUTOMOTIVE	0.0	0
2016-01-01	1946076	2016-01-01	13	AUTOMOTIVE	0.0	0
...	...	...	...	...	...	...
2016-12-31	2596209	2016-12-31	54	AUTOMOTIVE	18.0	0
2016-12-31	2596242	2016-12-31	6	AUTOMOTIVE	6.0	0
2016-12-31	2596275	2016-12-31	7	AUTOMOTIVE	5.0	0
2016-12-31	2596308	2016-12-31	8	AUTOMOTIVE	1.0	0
2016-12-31	2596341	2016-12-31	9	AUTOMOTIVE	5.0	0

19710 rows × 6 columns

In [14]:
```
#1.8 Count the number of zero values in the 'sales' column
zero_count = (data_train['sales'] == 0).sum()
total_count = data_train['sales'].count()
print("Number of zero values in the 'sales' column:", zero_count, "\n Number of total
```

```
Number of zero values in the 'sales' column: 119922
 Number of total values: 650430
 Percentage: 18.437341451040083 %
```

In [15]:
```
#Plot outliers for automotive
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 6))
sns.histplot(automotive_train['sales'], kde=True)
plt.title('Histogram for Outlier Detection')
plt.show()
```